

POWER BI 30 DAYS COURESE

Day 2 – Import Data into Power BI

Import Data in Power BI

Importing data means bringing data from files, databases, or online sources into Power BI for analysis and reporting.

Three Ways to Import Data in Power BI

1. Get Data Option

The first way is using the Get Data button from the Home tab.

Steps:

1. Click Get Data
2. Choose a data source
3. Select the file
4. Click Load (if data is cleaned) or Click Transform Data (if data need cleaning)

Use:

Used to import:

- Excel files
 - CSV files
 - SQL Server data
 - Online data
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2. Import Data from Excel Tile

The second way is clicking the Import data from Excel option on the canvas.

Steps:

1. Click Import data from Excel
2. Select the Excel file
3. Click Open

4. Data loads into Power BI Load (if data is cleaned) or Click Transform Data (if data need cleaning)

Use:

Quick and easy method for Excel files.

3. Data Pane / Fields Pane

The third way is using the Data Pane on the right side.

Steps:

1. Click the Data/Fields pane
2. Choose Get Data
3. Select the file or database
4. Load the data Load (if data is cleaned) or Click Transform Data (if data need cleaning)

Use:

Helps manage imported tables and fields.

ETL Tool

Full Form of ETL

E → Extract

T → Transform

L → Load

ETL is a process used to collect, clean, and load data into a system for reporting and analysis.

Or

ETL is the process of extracting data, transforming it into a useful format, and loading it into a system for analysis.

Steps of ETL

1. Extract

Data is collected from different sources.

Sources:

- **Excel**
- **Database**
- **Websites**
- **Cloud services**

Example:

Import sales data from Excel.

2. Transform

Data is cleaned and modified.

Tasks:

- **Remove duplicates**
- **Fix errors**
- **Change data format**
- **Filter unwanted data**

Example:

Remove blank rows from sales data.

3. Load

Cleaned data is loaded into a system like Power BI or a database.

Example:

Load transformed data into Power BI dashboard.

ETL in Power BI

Power BI uses Power Query as an ETL tool.

Power Query Helps To:

- **Import data**
- **Clean data**

- Transform data
- Load data into reports

Power Query

Power Query is a tool in Power BI used to:

- Import data
- Clean data
- Transform data

before creating reports and dashboards.

Or

Power Query is an ETL tool in Power BI used to import, clean, transform, and load data for analysis and reporting.

Uses of Power Query

1. Import Data

Power Query can import data from:

- Excel
- CSV
- SQL Server
- Web
- Cloud services

2. Clean Data

Power Query helps clean incorrect or unwanted data.

Example:

- Remove blank rows
- Remove duplicates
- Fix errors

3. Transform Data

Power Query changes data into a useful format.

Example:

- **Change column names**
- **Split columns**
- **Merge tables**
- **Filter data**

4. Load Data

After cleaning and transforming, data is loaded into Power BI.

Features of Power Query

- **Easy to use**
- **No coding required**
- **Step-by-step data transformation**
- **Saves transformation steps automatically**

Common Data Sources in Power BI

Power BI can connect to different types of data sources to create reports and dashboards.

1. Excel Workbook

Used to import Excel files into Power BI.

Example:

- **Sales data**
- **Employee details**
- **Student marks**

Use:

Helps analyze Excel data using charts and reports.

2. Power BI Semantic Models

Used to connect with existing Power BI datasets.

Use:

Reuse already published data and reports without creating data again.

3. Dataflows

Dataflows are cloud-based data storage and transformation tools.

Use:

- **Clean data**
 - **Transform data**
 - **Prepare data before reporting**
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4. Dataverse

Dataverse is Microsoft's secure cloud database.

Use:

Store and manage business application data safely.

Example:

Customer and company records.

5. SQL Server

Used to connect Power BI with SQL databases.

Use:

Import large amounts of business data from servers.

Example:

Company sales database.

6. Analysis Services

Used for advanced business analysis and data models.

Use:

Create enterprise-level analytics and reporting solutions.

7. Text/CSV

Used to import text files and CSV files.

Example:

- **Customer lists**
 - **Sales reports**
 - **Comma-separated data files**
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8. Web

Used to import data from websites or online pages.

Example:

- **Online tables**
 - **Web reports**
 - **Website data**
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9. OData Feed

Used to connect with online services and APIs.

Use:

Import live online data into Power BI.

10. Blank Query

Used to create custom queries manually in Power Query.

Use:

Advanced data cleaning and transformation.

11. More...

Shows many additional data source options.

Examples:

- **Google Analytics**
- **Oracle**
- **SharePoint**
- **MySQL**
- **PostgreSQL**

Use:

Connect Power BI with many different platforms and databases.

Note:

Workbook for Practise (on Data Workbook For Practise -> day 2 folder)